CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A mixture of iron filings, common salt, and fine sand is given. Which sequence is the most efficient? 1

(a) magnet, add water and filter, evaporate filtrate
(b) add water, evaporate, magnet
(c) filter, magnet, sieve
(d) evaporate first, filter, magnet

2. A candle burns, melts, gives out waste gases and its flame responds to wind. Why is it still non-living? 1

(a) it melts **(b)** it does not grow or reproduce on its own
(c) it is hot **(d)** it makes light

3. A student collected 10 mL of gas from a jar over water after a candle burnt out. The candle extinguished because: 1

(a) no gas remained **(b)** all nitrogen was used
(c) the jar broke **(d)** most oxygen was used

4. If sunlight takes 8 minutes to reach the Earth, the Sun's distance in terms of light travel time is: 1

(a) 8 light-minutes **(b)** 8 light-years **(c)** 8 light-hours **(d)** 8 light-seconds

5. 100 g of a mixture contains 20 g of iron, 30 g of sand and 50 g of salt. After separation, the mass of sand obtained is expected to be: 1

(a) 30 g **(b)** 20 g **(c)** 50 g **(d)** 100 g

6. Out of a tree, a bird and a pebble, which one is non-living? 1

(a) tree **(b)** pebble
(c) bird **(d)** both bird and tree

7. Which of these actions improves soil fertility naturally? 1

(a) adding compost **(b)** removing worms **(c)** burning plastics **(d)** adding plastic

8. Mass of an object is 60 kg on Earth. On the Moon (gravity one-sixth), its mass is: 1

(a) 360 kg **(b)** 10 kg **(c)** 6 kg **(d)** 60 kg

9. Why is alum added during water purification before filtration? 1

(a) it adds taste **(b)** it makes tiny particles clump and settle
(c) it kills all germs **(d)** it dissolves salt

10. A dry seed lies unchanged in a packet for months but sprouts after being watered. This shows that: 1

(a) sprouting is random
(b) seeds are non-living
(c) water makes seeds alive from stone
(d) a living thing can stay inactive until conditions suit it

11. A town has 1,00,000 vehicles each emitting smoke. Which step reduces the problem most? 1

(a) more cars **(b)** more public transport
(c) less trees **(d)** more roads only

12. Which of the following explains day and night on Earth? 1

(a) Earth revolves around the Sun once a day **(b)** Earth rotates on its axis
(c) the Sun moves around the Earth **(d)** the Moon blocks the Sun daily

13. In the mixture above, after the first step the residue on the filter paper is: 1

(a) sand **(b)** water **(c)** iron **(d)** salt

14. A student observes a crystal growing in a saturated solution. Which statement is true? 1

(a) it is living
(b) the crystal grows by outside deposition, not from within
(c) it respires
(d) it reproduces

15. In that experiment, water rises in the jar by about one fifth because: 1

(a) oxygen forms about one-fifth of air **(b)** water is heavy
(c) nitrogen forms one-fifth **(d)** jar is wide

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A mixture of 50 g salt and 50 g sand is given. After separation, 49 g of salt and 50 g of sand are recovered. Explain the 1 g loss. 2

17. A student claims that a river is living because it flows and changes shape. Rebut the claim. 2

18. Why does the water rise about one fifth in a jar when a burning candle goes out over water? 2

19. On a map of the solar system, 1 cm shows 150 million km. If the Sun-Earth distance is 150 million km, how far away is Mars at about 228 million km? 2

20. How would you separate a mixture of sugar crystals and wheat grains? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A mixture of salt, sand, chalk powder, iron filings and wooden chips needs to be separated. Suggest the best sequence. 3

22. A student claims that mushrooms are plants. Give three reasons why they are not. 3

23. Calculate the oxygen present in a room of air volume 50 m^3, assuming 21 per cent oxygen. 3

24. A day on planet X is 10 hours and its year is 5000 hours. How many days are there in its year? 3

OR

Describe why a mixture of water and kerosene separates into two layers. 3

25. Why can't we say that a seed is dead because it does not respire visibly? 3

OR

A school discards 200 sheets of paper daily. If both sides were used, how many sheets would be needed? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two beakers contain salt water and muddy water. Explain with the help of experiments how you would show that one has dissolved salt and the other has insoluble mud. 5

OR

Design an investigation to show that seeds need water, air and suitable temperature to germinate. Describe set-ups and results. 5

27. Design a survey to find the use of natural resources in your home for a week and suggest five ways to cut it. 5

OR

Design a model of the solar system for your school science fair describing the materials, scales and observations. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND